

Trees

Introduction

- A tree is a connected undirected graph with no simple circuits.
- The family tree of the male members of the Bernoulli family of Swiss mathematicians is shown in Figure 1.

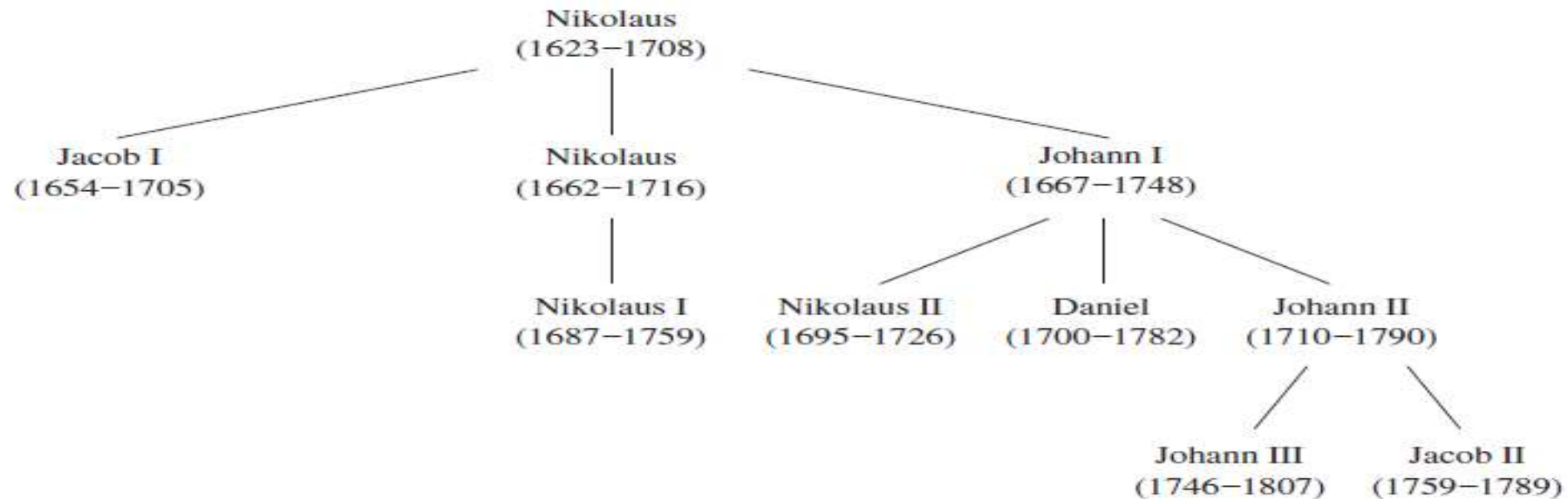
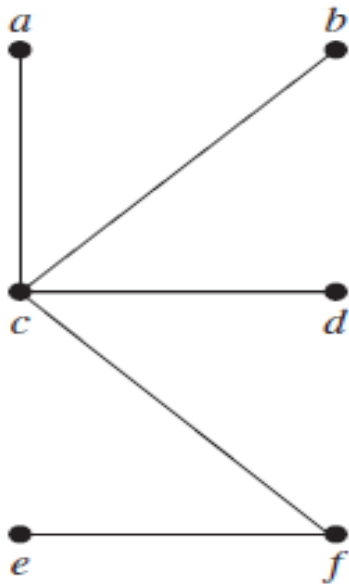
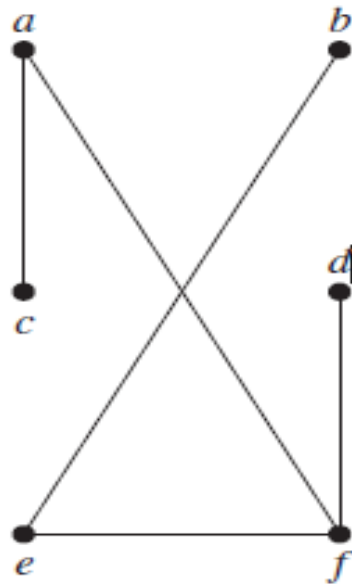


FIGURE 1 The Bernoulli Family of Mathematicians.

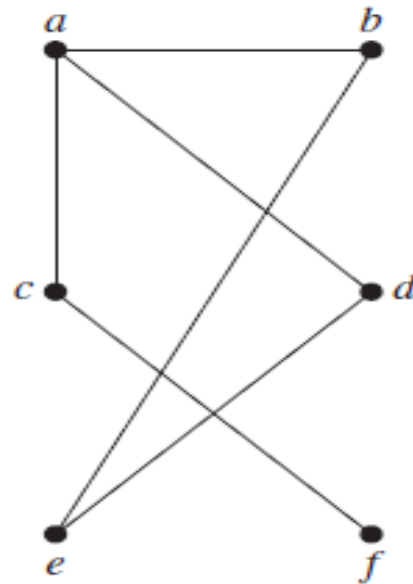
- Tree cannot have a simple circuit, a tree cannot contain multiple edges or loops. Therefore any tree must be a simple graph.
- Which of the graphs shown in Figure are trees?



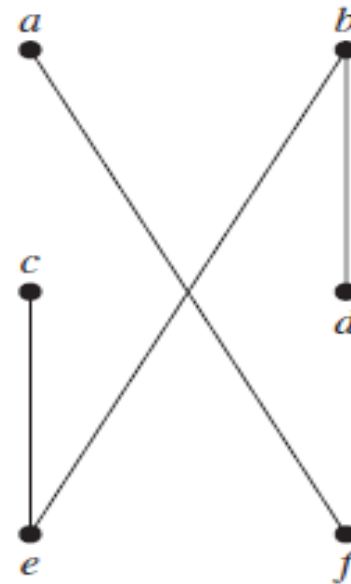
G_1



G_2

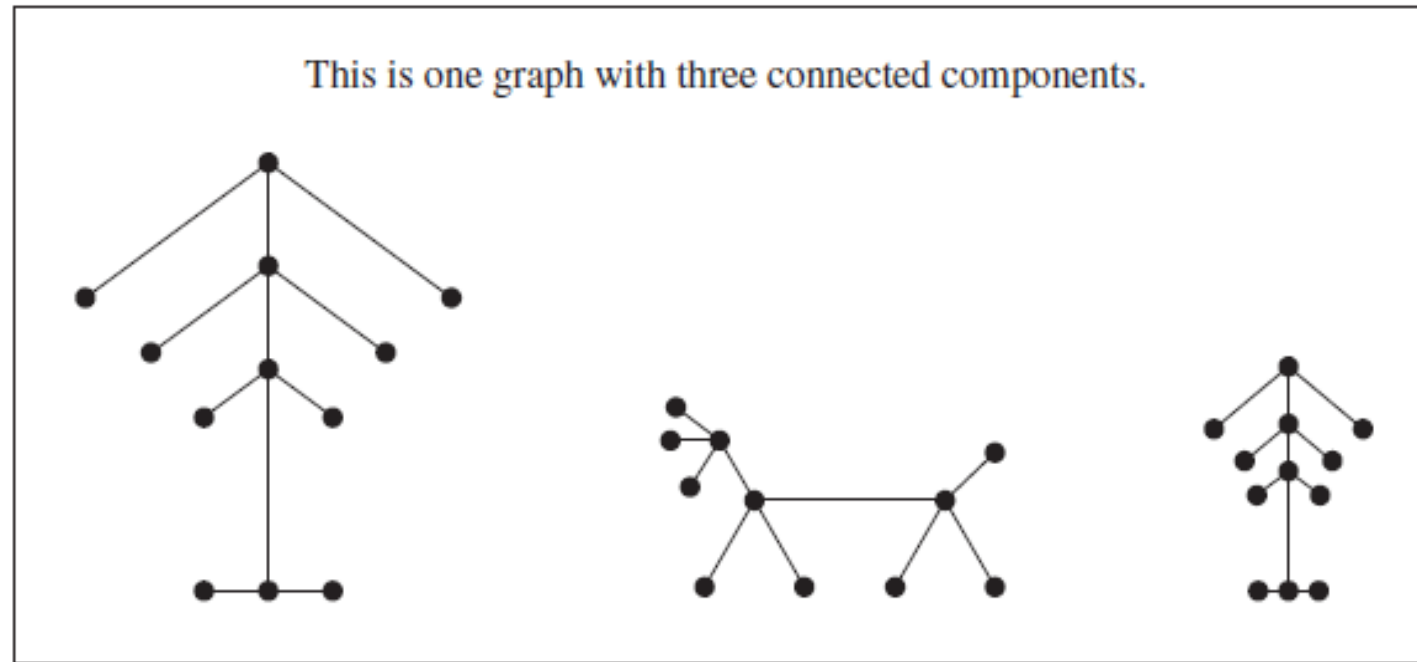


G_3



G_4

- G_1 and G_2 are trees, because both are connected graphs with no simple circuits. G_3 is not a tree because e, b, a, d, e is a simple circuit in this graph. Finally, G_4 is not a tree because it is not connected.
- Graphs are called **forests** which have the property that each of their connected components is a tree.



- **Theorem: An undirected graph is a tree if and only if there is a unique simple path between any two of its vertices.**

Proof: Assume that T is tree. Since T is a Tree, it is a connected simple graph with no simple circuits.

Let x and y be the two vertices of T . We know that every connected graph has a simple path between every pair of vertices. So there is a simple path between x and y .

This path must be unique because, if the path between x and y is not unique then there is another path between x and y that uses a edges different from the path between x and y for first path , then reversing the path i.e. going from x to y from the first path and going from y to x through second path forms a circuit.

This is a contradiction that T is a tree; hence there is unique simple path between any two vertices of a tree.

- Again, assume that there is a unique simple path between any two vertices of a graph, say T .
- Since there is a path between any two vertices of a graph, the graph is connected.
- Now, we can show that the graph T cannot have simple circuit.
- Had there been a simple circuit, there would be two simple paths between two vertices, say x and y , and the two simple paths between x and y would create a simple circuit where first path goes from x to y and the second path goes from y to x .
- This violates our assumption that the path is unique.
- Hence, a graph with a unique simple path between any two vertices is tree.

- A *rooted tree* is a tree in which one vertex has been designated as the root and every edge is directed away from the root.

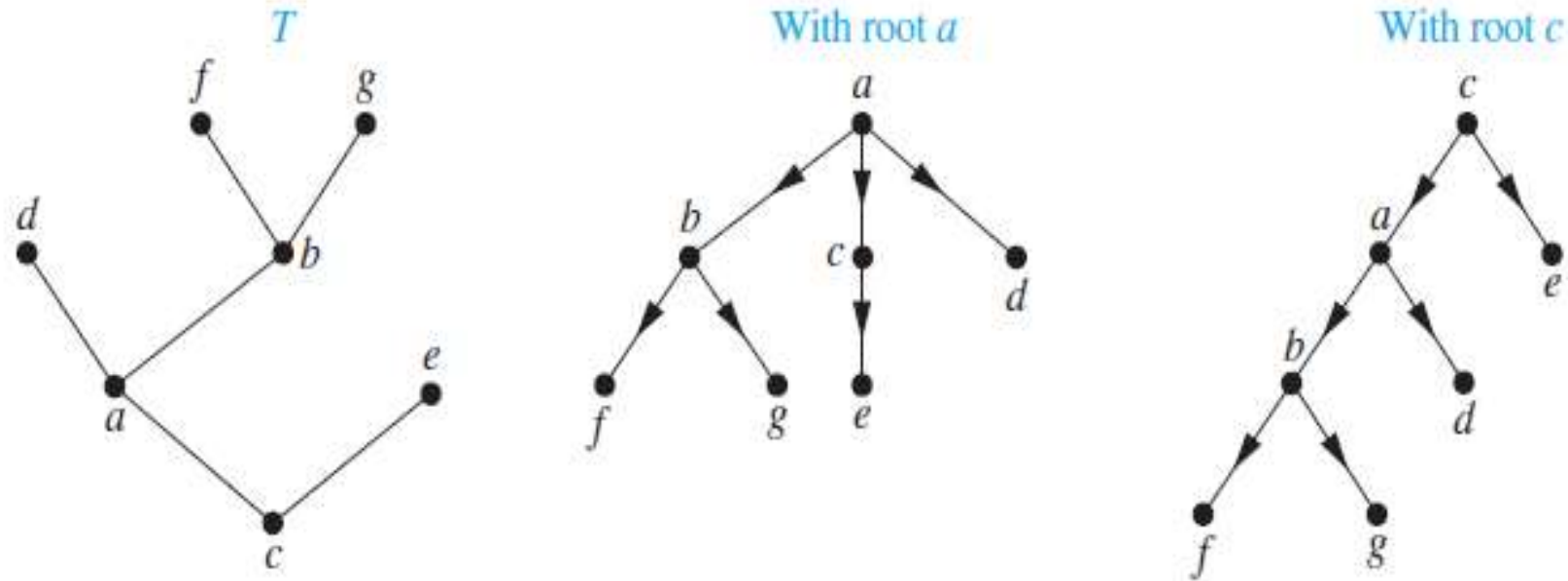


FIGURE 4 A Tree and Rooted Trees Formed by Designating Two Different Roots.

- We can change an un-rooted tree into a rooted by choosing any vertex as root. The tree in which root is defined produces a directed graph as shown in above figure.
- Take a root T . if v is the vertex in T other than root, then **parent** of v is vertex u in T such that there is a directed edge from u to v .
- In this scenario, v is called **child** of u .
- Vertices with same parents are called **siblings**.
- All the vertices that appear in the path from root to some vertex v in T , including root are called **ancestors** of v .
- The **descendants** of vertex v are those vertices that have v as their ancestor.
- All the vertices that have children are called **internal vertices** (root is also an internal vertex if the tree has more than one vertices).
- A vertex of a rooted tree is called a **leaf** if it has no children.
- If a is a vertex in a tree, the **subtree** with a as its root is the subgraph of the tree consisting of a and its descendants and all edges incident to these descendants.

- **EXAMPLE:** In the rooted tree T (with root a) shown in Figure 5, find the parent of c , the children of g , the siblings of h , all ancestors of e , all descendants of b , all internal vertices, and all leaves. What is the subtree rooted at g ?

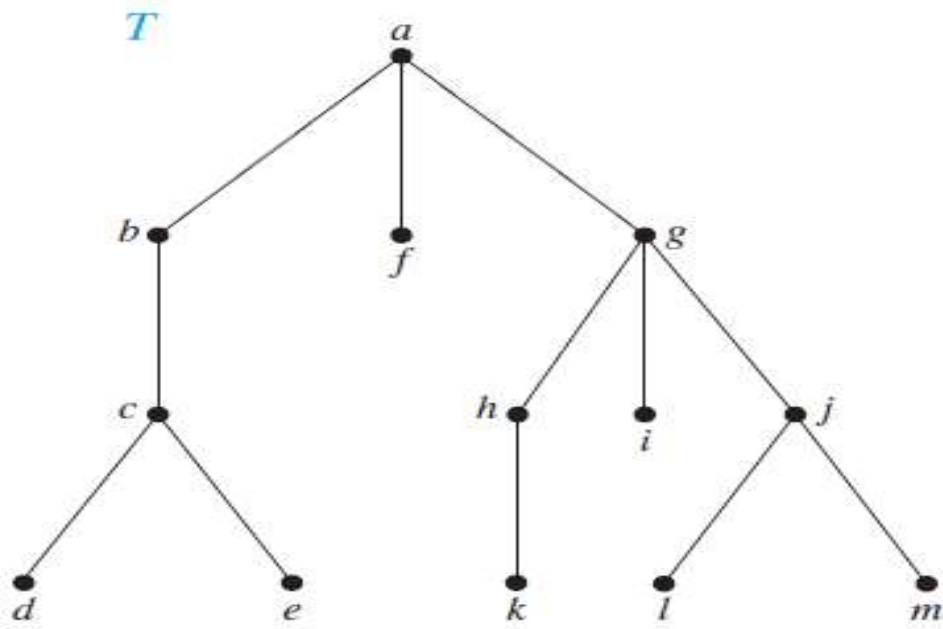


FIGURE 5 A Rooted Tree T .

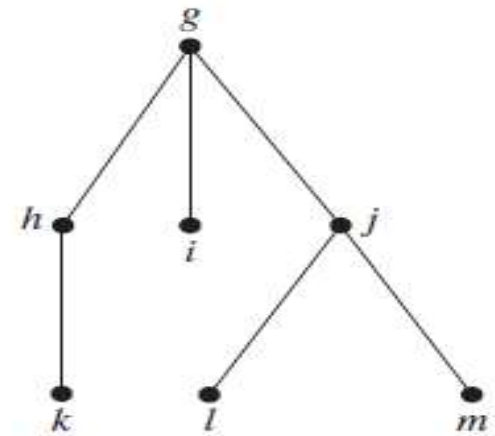


FIGURE 6 The Subtree Rooted at g .

- The parent of c is b . The children of g are h , i , and j . The siblings of h are i and j . The ancestors of e are c , b , and a . The descendants of b are c , d , and e . The internal vertices are a , b , c , g , h , and j . The leaves are d , e , f , i , k , l , and m . The subtree rooted at g is shown in Figure 6.
- A **m-ary** tree is a rooted tree in which every internal vertex has no more than m children. It is called full **m-ary** tree if every internal vertex has exactly m children.
- Example: Binary tree i.e. 2-ary tree.

- Are the rooted trees in Figure 7 full m -ary trees for some positive integer m ?

Ans: T_1 is a full binary tree because each of its internal vertices has two children. T_2 is a full 3-ary tree because each of its internal vertices has three children. In T_3 each internal vertex has five children, so T_3 is a full 5-ary tree. T_4 is not a full m -ary tree for any m because some of its internal vertices have two children and others have three children.

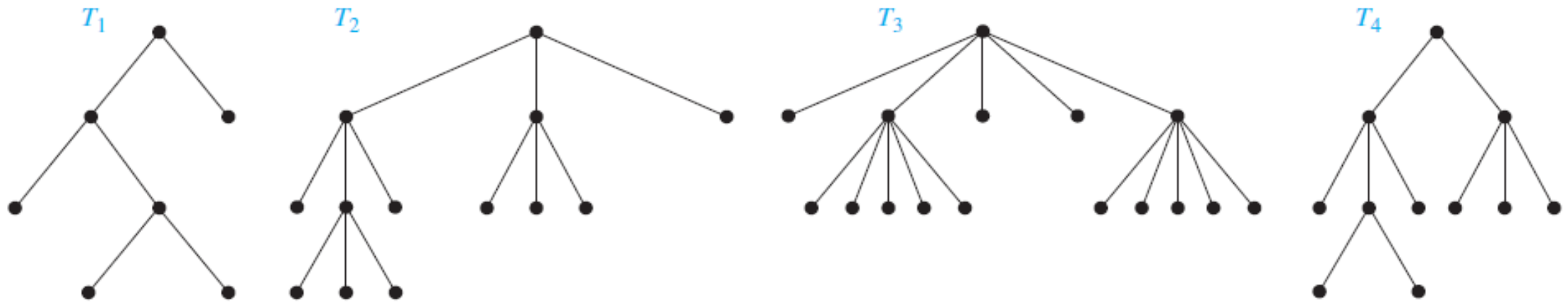


FIGURE 7 Four Rooted Trees.

- **Ordered Rooted Tree:** An ordered rooted tree is a rooted tree where the children of each internal vertex are ordered. For e.g. in ordered binary tree (also called just a binary tree) if an internal vertex has two children then the first child is called **left child** and the second child is called **right child**.
- What are the left and right children of d in the binary tree T shown in Figure 8(a) [left children= f and right children= g] (where the order is that implied by the drawing)? What are the left and right subtrees of c ?

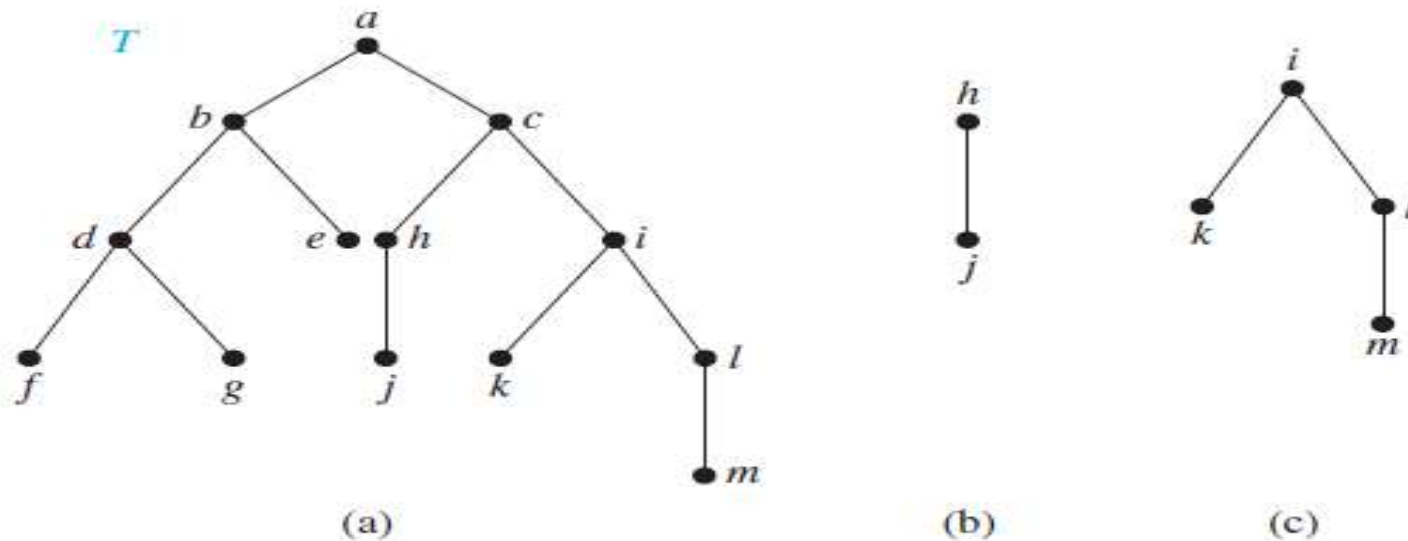


FIGURE 8 A Binary Tree T and Left and Right Subtrees of the Vertex c .

- An ordered rooted tree is a rooted tree where the children of each internal vertex are ordered from left to right.
- In an ordered binary tree (usually called just a binary tree), if an internal vertex has two children then the first child is called the left child and the second child is called the right child.
- The tree rooted at the left child of vertex is called the left subtree of this vertex, and the tree rooted at the right child of a vertex is called the right subtree of the vertex.

- Tree as models: **Computer File Systems**
- Files in computer memory can be organized into directories.
- A directory can contain both files and subdirectories. The root directory contains the entire file system. Thus, a file system may be represented by a rooted tree, where the root represents the root directory, internal vertices represent subdirectories, and leaves represent ordinary files or empty directories.
- One such file system is shown in Figure 11. In this system, the file khr is in the directory rje. (Note that links to files where the same file may have more than one pathname can lead to circuits in computer file systems.)

Computer File System

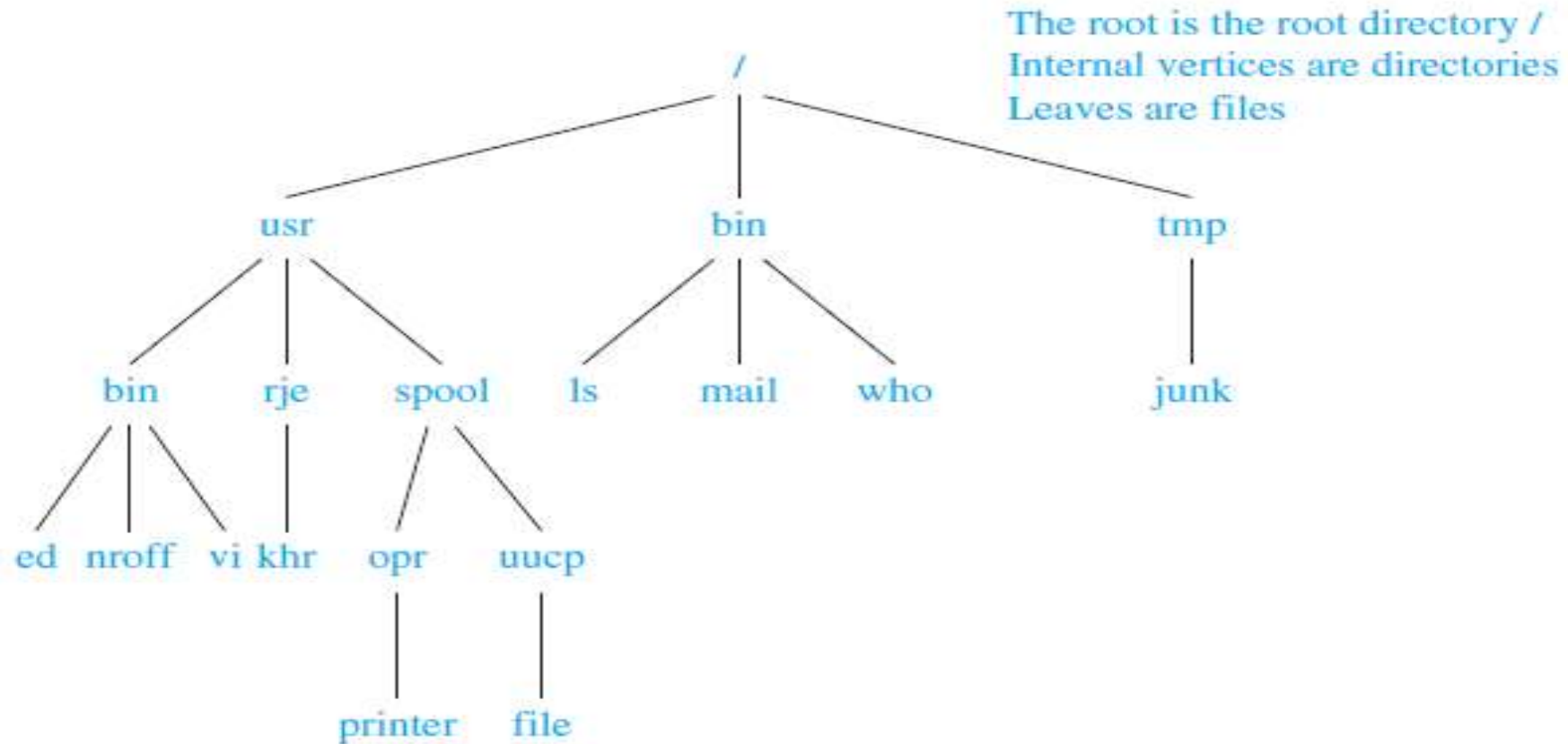


FIGURE 11 A Computer File System.

- **Tree Connected Parallel Processor:** A **tree-connected network** is another important way to interconnect processors. The graph representing such a network is a complete binary tree, that is, a full binary tree where every root is at the same level. Such a network interconnects $n = 2^k - 1$ processors, where k is a positive integer.
- A processor represented by the vertex v that is not a root or a leaf has three two-way connections—one to the processor represented by the parent of v and two to the processors represented by the two children of v . The processor represented by the root has two two-way connections to the processors represented by its two children. A processor represented by a leaf v has a single two-way connection to the parent of v .
- We display a tree-connected network with seven processors in Figure 12

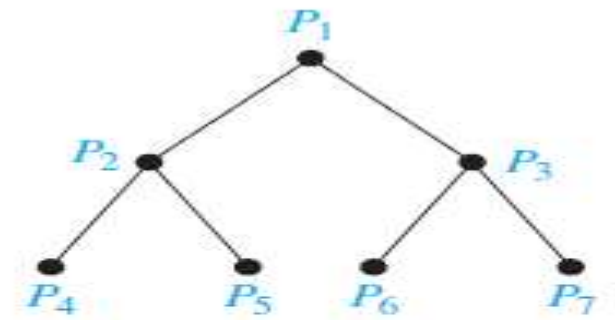


FIGURE 12 A Tree-Connected Network of Seven Processors.

- We now illustrate how a tree-connected network can be used for parallel computation. In particular, we show how the processors in Figure 12 can be used to add eight numbers, using three steps.
- In the first step, we add x_1 and x_2 using P_4 , x_3 and x_4 using P_5 , x_5 and x_6 using P_6 , and x_7 and x_8 using P_7 .
- In the second step, we add $x_1 + x_2$ and $x_3 + x_4$ using P_2 and $x_5 + x_6$ and $x_7 + x_8$ using P_3 . Finally, in the third step, we add $x_1 + x_2 + x_3 + x_4$ and $x_5 + x_6 + x_7 + x_8$ using P_1 .
- The three steps used to add eight numbers compares favorably to the seven steps required to add eight numbers serially, where the steps are the addition of one number to the sum of the previous numbers in the list.

Properties of trees (Important)

Theorem: A tree with n vertices has $n-1$ edges.

Proof: Here we use mathematical induction to prove this theorem.

Basis Step: When $n=1$, a tree with $n=1$ vertex has no edges. It follows that the theorem is true for $n=1$.

Inductive Hypothesis: Assume that the tree with k vertices has $k-1$ edges, where k is positive integer.

Inductive step: Suppose that a tree T has $k+1$ vertices and that v is a leaf of T . Removing the vertex v and the associated edge from T produces a tree T_1 with k vertices, since the resulting graph is still connected and has no simple circuits.

By the induction hypothesis, T_1 has $k-1$ edges. Hence, T has k edges since it has one more edge than T_1 , the edge connecting v to its parent.

Theorem: A full m-ary tree with i internal vertices contain $n=mi+1$ vertices.

- Proof: In a full m-ary tree all the vertices but the root are child of the internal vertices. Since each internal vertex has m children there are mi vertices in the tree other than the root. Hence the tree has $n = mi + 1$ vertices.

➤ The level(depth) of a vertex v in rooted tree is the length of the unique path from the root to vertex. The level of root is zero. The height of the rooted tree is the length of the longest path from root to any vertex.